

Abu Sufiun Rhyme

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SUMMARY

Data Analyst and Data Scientist with a background in Applied Statistics and Data Science and hands-on experience in data analysis, statistical modeling, machine learning, and causal inference. Skilled in R, Python, SQL, Excel, Power BI and STATA. Developed practical projects including a credit risk prediction system, interactive sales analytics dashboard, and causal machine learning R package. Experienced in transforming complex datasets into meaningful insights and data-driven solutions for research and business applications.

EDUCATION

Institute of Applied Statistics and Data Science (Formerly ISRT), University of Dhaka Dhaka, Bangladesh
Master of Science (M.Sc.) in Applied Statistics and Data Science Dec. 2025 – Present

- **Relevant Coursework:** Causal Inference, Machine Learning, Big Data Analytics, Bayesian Statistics, Longitudinal Data Analysis, Statistical Signal Processing, Spatial Statistics, Python

Institute of Applied Statistics and Data Science (Formerly ISRT), University of Dhaka Dhaka, Bangladesh
Bachelor of Science (B.Sc.) in Applied Statistics Jan. 2022 – Dec. 2025

- **Relevant Coursework:** Regression Analysis, Time Series Analysis, Statistical Inference, Research Methodology, Multivariate Analysis, PCA, FA, Clustering, Classification, Survival Analysis, Sampling Design, Epidemiology, Population Studies & Demography, R, SQL
- **Result:** 3.75 / 4.00

Amrita Lal Dey College Barishal, Bangladesh
Higher Secondary Certificate (HSC) in Science Jun. 2018 – Jan. 2021

- **Result:** 5.00 / 5.00

Alipura Union Secondary School Patuakhali, Bangladesh
Secondary School Certificate (SSC) in Science Jan. 2016 – May 2018

- **Result:** 5.00 / 5.00

TECHNICAL SKILLS

Programming Languages : R, Python, SQL

Data Analysis & Statistical Computing : tidyverse, dplyr, NumPy, pandas, SciPy, statsmodels

Machine Learning : scikit-learn, Decision Tree, XGBoost, Random Forest, Gradient Boosting, SVM, KNN

Data Visualization & Business Intelligence : Microsoft Power BI, Looker Studio, Shiny, ggplot2, matplotlib, seaborn

Databases & Data Warehousing : PostgreSQL, Google BigQuery, MySQL

Tools : Microsoft Excel (Advanced), Google Sheets (Advanced), STATA, SPSS, LaTeX, R Markdown

RESEARCH EXPERIENCE

MS Thesis (In Progress)  Dec. 2025 – Present

Estimating the Causal Effect of Higher Education on Antenatal Care (ANC) Utilization with Sensitivity Analysis Using Machine Learning Methods

- Estimated the causal effect of maternal higher education on ANC utilization using BDHS 2022 data.
- Applied BART, Causal Forests, Causal Additive Forests, Causal MARS, and Causal Boosting for causal effect estimation.
- Used Parametric Standardization as a benchmark and evaluated methods through simulation studies.

- Conducted sensitivity analysis for unmeasured confounding and evaluated heterogeneous treatment effects using CATE estimation.

BS Project

Oct. 2024 – May 2025

Assessing the Risk Factors of Under-Five Child Mortality: A Study Based on BDHS 2022

- Analyzed 8,800+ observations from the Bangladesh Demographic and Health Survey (BDHS 2022) using R and STATA.
- Applied multilevel logistic regression, hypothesis testing, and exploratory data analysis to identify key determinants of under-five mortality.
- Developed predictive models with excellent discrimination performance (AUC = 0.91).
- Created statistical visualizations and reports to support evidence based decision making.

SELECTED PROJECTS

causalML – R Package

Causal Machine Learning and Sensitivity Analysis

- Developed an R package for causal effect estimation using Causal Forest, BART, Causal Boosting, Causal MARS, and Causal Additive Forest methods.
- Implemented simulation tools, model comparison, ATE/CATE estimation, and sensitivity analysis functionality.

LoanGuard – Streamlit

Credit Risk Prediction System

- Developed an interactive Python Streamlit application for credit risk analysis, loan default prediction, exploratory data analysis, and customer risk assessment.
- Implemented and compared multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, SVM, and KNN.
- Built an end-to-end machine learning pipeline with data preprocessing, feature encoding, scaling, model evaluation, and real-time new customer prediction.

Sales BI Dashboard – Shiny

Interactive Sales Analytics and Business Intelligence Dashboard

- Developed an interactive Business Intelligence dashboard using R Shiny for analyzing sales performance, business KPIs, products, regions, customers, and salespersons.
- Implemented interactive filters, KPI cards, sales trend analysis, product and regional performance analysis, customer insights, and transaction level data exploration.
- Integrated automated report and visualization download features to support business reporting and decision making.

ONLINE COURSES & CERTIFICATIONS

- **Data Analysis with Python** freeCodeCamp (Dec. 2025) 
- **Data Science Certificate Program** Ostad (Feb. 2025) 
- **Data Analyst Job-Ready Bootcamp** Datasolution-360 (Aug. 2025) 
- **SQL for Data Analysis** Dataskool (Dec. 2023)

REFERENCES

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